

INTRODUCTION TO INTERACTION DESIGN

The crux of the idea is that, whenever we design an application, we need to consider the perspective of the end-user and make the relationship between the user and app meaningful. Try to keep the options as simple as possible for the end-user.

When developing a product, you need to consider which users will be utilising your product, and *involve* them in your product by having them test a prototype and give feedback.

Ergonomics: the usability

Proof of good design is evident when people look at an item and intuitively know how to use it.

Four basic activities of interaction design: can be very elaborate, but hey - at least we get user feedback from the start!

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EVALUATION

TOBII GLASSES MOBILE EYE: when you give them to a user and ask them to test, say, a mobile application, you can track how they use it. Via this, you can figure out what functionality they struggle with, what they're most interested in...etc.

Problems and actions:

Lack of affordance: affordance is when a user knows how to use a device based on prior experience with similar devices. Despite the users having used iPhones before, they struggled with getting around an iPad.

Usability testing conditions:

- Tasks should only take 30 minutes or it would affect the results as no one has that much time.
Think about which aspect of the application you need to focus on the most, and give that to the user.

Informed consent: create a separate, specific form, easy to use. Give it to the user to gather all required information.

FIELD STUDIES

One field where field studies are very popular is in agriculture, where studies are used to develop hardware or software.

THE THREE QUESTIONS

- Will the correct action be sufficiently evident to the user?: AFFORDANCE comes in again.
Will a user know what a certain icon does when they see it?

PREDICTIVE MODELS

- Niche area.
- Ask a GenAI tool to comment on the usability of an interface using these key aspects.
- You can even eliminate users completely!
- Usability testing these days involves GenAI or agents.

FITTS' LAW

- The larger the icon, the easier it is going to be for the end-user to find and interact with it.

Where should an icon be placed?

You need to make your own notes about what kind of evaluation you're going to use for your application.

Ten Usability Heuristics

1. When the user clicks on an icon, or performs an action that takes some time, show a pop-up that shows something is happening.
2. The design should speak the users' language, not yours.
3. The design should allow users to rectify the mistakes they've made.
4. There are certain conventions that need to be followed. For instance, when someone opens a website, they expect to see the landing page. Certain conventions exist by default, and these are the ones that need to be followed.
5. People will complain if they see a lot of errors, as this influences user experience negatively.
6. k
7. When we create applications, we need to have shortcuts in place so experienced users can use these shortcuts. Novice users can follow the formal path. Hence, we need to have two versions of our application running side-by-side: one for the everyday user, one for the experts. It would be optimal if these two could be merged together.
8. Do not put any unhelpful information on the interface, and maintain this consistency across all units of the interface.
9. Create error messages in plain English, so users can understand.
10. Whether it's for consistency or affordance, we usually try to avoid creating a manual for apps. In certain instances, a manual would be necessary. But generally, the ambience of the app should allow users to find their way around it without the need for a manual.

FIVE VISUAL-DESIGN PRINCIPLES IN UX

1. Balance: have an imaginary axis, and ensure the colour scheme isn't too intense on one side, or too light on a side, or it's going to be visually challenging.
2. Scale: make sure the scale actually makes sense. It's not too big, it's not too small, it's not too challenging. With a vertical list, for instance, chances are high that we might click the wrong option because all the items are tightly compact together.
3. Contrast
4. Visual hierarchy: Something very important should be at the top. After that, in order of decreasing importance or hues, the others should follow.
5. Gestalt principles:

TEST PLAN

As a bare minimum, you need to create one test.

INTERFACES

- Mega menu:
 - The size of the menu is so gigantic, a mobile phone user would not be able to see the whole thing.
 - You don't have to search throughout the website for a specific item.

SELECTING A COUNTRY FROM A SCROLLING WINDOW

- You need to scroll all the way down to the bottom to find your country.

WHAT DOES THIS ICON MEAN?

Crop and rotate (better: polygon cartwheel)

USABILITY VERSUS AESTHETICS

It's all about getting the relevant information across to users. Compare NSBM's landing page with that of any UK university. You'll see that the former is so cluttered with information, whereas the latter is simpler.

PROTOTYPING

3D printing is an example of prototyping.

COMPROMISES IN PROTOTYPING

You need to identify compromises and let customers know. For instance, if the prototype is

slow, inform the customer of this beforehand. Don't give them all the features, but ensure that the selection you give are functional.

SCENARIOS:

- Short stories about how a user interacts with an application.
- Think about everything from the end-user's perspective.
- Motivations for using the app in the first place: what they're trying to achieve.
- Scenarios are not equal to user stories, because they can have multiple user stories in them, or relate to multiple user stories.

CONCRETE DESIGN

Indigenous knowledge: get opinions from a person educated in that field.

SDK

- Simplify the developers' tasks.
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UxD IN THE SDLC

LEANUX

- Research and learning: requirements and what we want to deliver.
- If you don't have time to interact with customers, you can be the customer yourself. Yes, you can play around with imaginary customers.

DOCUMENTATION

- Agile requires minimal documentation, so you may get rid of all the documentation, or minimise it and make it as brief and succinct as possible. The more documentation there is, the more time is wasted.

A SIMPLE INTERACTION DESIGN LIFECYCLE MODEL

- Designing alternatives: if you think a feature is necessary but you don't have the required skills to implement it, you may design alternative ways of implementing it.
- Prototyping: this is where the low-fidelity and high-fidelity prototypes come in. We usually begin with low-fidelity prototypes, which include sketches.

WHO ARE THE STAKEHOLDERS?

- Manufacturers., stores that will use the app.

WHAT ARE THE USERS' NEEDS?

- Users rarely know what can be done with an app. Hence, as the developer, explain it to them.

VOLERE SHELL

- This template is available if you want to focus on one particular requirement from a category, so you can benchmark it.
- Source: who provided you with information.

USABLE SECURITY

- Often, a lot of information is offered about how to choose a password. It's useful from a security perspective, but a nightmare for the end user.
- Think about how you can reduce the amount of effort people have to put in - even for something as simple as logging in.

USING PROBES TO ENGAGE WITH USERS

- When interacting with users, you can use simple questions. There are different ways in which you can probe - or rather, prompt - users to answer the questions.
- Asking for the user's age might throw them off. What you can do, is to provide an age range.

PERSONAS

- When you create a persona, you need to consider the pros and the cons.
- You can bring them to life with a name, characteristics...etc. You can explore their relation to the product.

VISUALISATION

It is about understanding users' requirements and describing them. Since we are moving away from documentation, we need a visual way of showing this.

INFORMATION VISUALISATION

- It can apply to groups of users, or one user who is building an app for themselves.

DATA TYPES

- After you've gathered information from the user, *how* are you going to present it? You may use visualisation - a graph, a chart...etc.
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DIGITAL IMAGE PROCESSING

Friday: mock test

DIGITAL IMAGE

- Alpha: controls the opacity/ transparency.

NOISE REMOVAL

- Noise is seen as white dots on an image that distorts it.

RELATIONSHIP

Image processing: taking a low-quality image, turning it into a digital image and putting it through an image processing algorithm.

Image analysis:

Computer vision: feeding an image into an AI-based system to identify features.

KEY STAGES IN DIGITAL IMAGE PROCESSING

Image enhancement: denoising the image

Morphological processing: identifying what the image is about.

Representation and description: identifying what the image is for.

In addition to the main steps in any image processing system, you also have applications that can resize or compress images.

LINE AND CIRCLE DRAWINGS

DDA

$\max(\text{abs}(dx), \text{abs}(dy)) \rightarrow$ maximum of ((absolute value of dx) or (absolute value of dy))

Why did we round x down?

We are limited to operating one pixel at a time. We won't change that, because that's the fundamental of computer science.

In this particular instance, our step was 8. Since dx was 4, we ended up getting an increment value of 0.5. But what if dx was 3? You'd end up getting a decimal value that was less than 0.5. However, the computer can only operate in whole pixels, so whatever value we get for the

increment is rounded off to its nearest whole number. *Which*, of course, is perhaps DDA's biggest limitation.

BRESENHAM'S LINE ALGORITHM

1. Start with two points the same way you did with DDA.

$$(x_1, y_1) =$$

$$(x_2, y_2) =$$

2. Calculate the gradient, but keep everything as an integer.

$$m = y_2 - y_1 /$$

CHALLENGES IN HCI

What is a prototype in interaction design? In interaction design, a prototype can be (among other things):

- A series of screen sketches
- A storyboard, for example, a cartoon-like series of scenes

Why prototype?

Team members can communicate effectively

Prototypes answer questions and support designers in choosing between alternatives

Storyboard is a low-fidelity prototype; Uses a medium which is unlike the final medium, for example, paper or cardboard

is a series of sketches showing how a user might progress through a task using the product

Often used with scenarios, bringing in more detail and a chance to role play